28/09/2026 **Experiment - 9 : End - End communication at transport layer**

(a) **Aim** : To implement echo client server using TCP/UDP packets.

Procedure :-

* Create a socket using Python socket programming.
* Specify the IP address & port number.
* Bind the server socket to the IP address and port.
* Make the server listed for client connections.
* Connect client to server.
* Send & receive message b/w client & server.
* Display received message.
* Close the socket connection.

Code :-

(i) TCP Server :-

```
import socket
server = socket.socket(socket.AF_INET,
                       socket.SOCK_STREAM)

server.bind("127.0.0.1", 5000)
server.listen(1)
print("Server is waiting for connection")
conn, addr = server.accept()
print("Client connected")

while True:
    data = conn.recv(1024).decode()
    if not data:
        break
    print("Client: ", data)
    conn.send(data.encode())
conn.close()

server.close()
```

(ii). TCP_receiver (client) :-

```
import socket
client = socket.socket (socket.AF_INET,
                        socket.SOCK_STREAM)
client.connect (("127.0.0.1", 5000))
message = input ("Enter message : ")
client.send (message.encode())
reply = client.recv (1024).decode()
print ("Server : ", reply)
client.close()
```

Terminal Server	Terminal Client
python tcp-server.py : server is waiting for connection. Client connected. Client : Hi	python tcp-client.py Enter message : Hi Server : Hi TCP Echo is working.

Result : The TCP/UDP packets using echo client server socket program is executed successfully.

(5) Aim : To implement chat client-server using TCP packet

Code :-

(i) Chat - server side

```
import socket
server = socket.socket (socket.AF_INET,
                        socket.SOCKET_STREAM)

server.bind (("127.0.0.1", 6000))
server.listen(1):
print ("Waiting for client....")
conn.addr = server.accept()
print ("Client connected")
while True :
    message = conn.recv (1024).decode()
    if message.lower() == "bye":
        print ("Client ended chat")
    break
    print ("Client ", message)
    reply = input ("Server")
    conn.send (reply.encode())
    if reply.lower() == "bye":
        conn.close()
```

(ii) Chat - client side :-

```
import socket
client = socket.socket (socket.AF_INET,
                        socket.sock_STREAM)
client.connect (("127.0.0.1", 6000))
print ("Connected to Server")
while True:
    message = input ("client")
    client.send (message.encode())
if message.lower() == "bye":
        break
reply = client.recv(1024).decode()
print ("Server", reply)
if reply.lower() == "bye":
    break
client.close()
```

Terminal 1	Terminal 2
python chat_server.py	python chat_client.py
Waiting for client	Connected to server
client connected	
Client : Hi	Client : Hi
Server : Hello	Server : Hello
Client : How are you?	Client : How are you?
Server : Fine	Server : Fine

Result : The chat programming using sockets by TCP client & server was implemented successfully.